WAI - Oefenzitting 1

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1 Useful MATLAB Commands

```
eig             Computes all eigenvalues and eigenvectors
eigs            Find a few eigenvalues and eigenvectors of a matrix
cond            Computes the condition number for inversion
condest         Computes 1-norm condition number estimate
norm            Vector and matrix norms
normest         2-norm estimate
minres          Solve linear system using the minimum residual method
gmres           Solve linear system using the generalized minimum residual method
ilu             Incomplete LU factorization
ichol           Incomplete Cholesky factorization
decomposition   Matrix decomposition for solving linear systems
plot            2-D line plot
semilogx        Semilog plot (x-axis has log scale)
semilogy        Semilog plot (y-axis has log scale)
subplot         Divides the current figure into an m-by-n grid
spy             Visualize sparsity pattern of matrix
sparse          Create sparse matrix
nnz             Number of nonzero matrix elements
whos            List variables in workspace, with sizes and types
tic             Start stopwatch timer
toc             Read elapsed time from stopwatch
help            Display help text in Command Window
```

2 Exercises

Exercise 1. Start with the matrix A provided at the link below: `https://sparse.tamu.edu/DIMACS10/delaunay_n15`. Perform the following tasks:

1. **Matrix properties.** Load the matrix A and check its main properties:
   - Is the matrix sparse?
   - Is the matrix symmetric?
   - How many nonzero elements does it contain?
   - What is the condition number of A?
2. **Arnoldi and Lanczos Methods.** Implement the following two functions:
   - `[lam, res, X, V, H] = arnoldi(A, x0, m)` — performs m iterations of the **Arnoldi method** for a matrix A.

- `[lam, res, X, V, T] = lanczos(A, x0, m)` — performs m iterations of the **Lanczos method** for a symmetric matrix A.

Input:

- `A` $\in \mathbb{R}^{n\times n}$: square matrix (symmetric for the Lanczos method);
- `x0` $\in \mathbb{R}^{n}$: initial vector;
- `m` $\in \mathbb{N}$: number of iterations (dimension of the Krylov subspace).

Output:

- `lam` $\in \mathbb{R}^{m}$: vector of the *Ritz values*, i.e. the approximate eigenvalues of A;
- `res` $\in \mathbb{R}^{m}$: vector containing the residual norms of each Ritz pair;
- `X` $\in \mathbb{R}^{n\times m}$: matrix whose columns are the approximate eigenvectors of A (the Ritz vectors);
- `V` $\in \mathbb{R}^{n\times (m+1)}$: orthonormal basis of the Krylov subspace;
- `H/T` $\in \mathbb{R}^{(m+1)\times m}$: upper Hessenberg or tridiagonal matrix obtained from the Arnoldi or Lanczos process, respectively.

Using these two functions, compute the first m Ritz values of the matrix A for

$$m \in \{5, 10, 30, 50, 100, 250, 300\}.$$

For each value of m, compute and record:

- the computational time required by the Arnoldi and Lanczos methods;
- the orthogonality of the basis vectors, measured as

$$\|V^{\mathsf{T}}V - I\|_2;$$

- the residual norms $\|r^{(m)}\|_2$ with $r_i^{(m)} = A\tilde{x}_i - \tilde{\lambda}_i\tilde{x}_i = \alpha_i v_{m+1}$, for $i = 1, \ldots, m$[1].

Finally, plot the results in a single figure using the `subplot` function:

- one subplot for the computational time,
- one subplot for the orthogonality error,
- one subplot for the average residual norm.

3. **Reorthogonalization.** Modify your Arnoldi and Lanczos implementations to include *full reorthogonalization* at each step. Repeat all computations and compare the results (time, orthogonality, and residuals) with the previous versions.

Exercise 2.

In this exercise, you will analyze and solve systems involving **two matrices**:

- **Matrix** A (order 2529): `http://sparse.tamu.edu/HB/orani678`
- **Matrix** B (order 1083): `https://sparse.tamu.edu/HB/bcsstk09`

1. **Matrix properties.** For each matrix A and B, examine its main properties:
   - Is the matrix sparse?

[1] $\alpha_i = h_{m+1,m}(e_m^{\mathsf{T}} s_i)$

- Is the matrix symmetric?
- How many nonzero elements does it contain?
- What is the condition number of the matrix?

2. **System solving.** Consider the linear systems

$$Ax = b, \quad b = \texttt{rand(2529,1)},$$

and

$$By = c, \quad c = \texttt{rand(1083,1)}.$$

For each system:

- **Method selection:** Choose an appropriate numerical method for solving the linear system based on the matrix properties (e.g., symmetric vs. non-symmetric).
- **Tolerance experiments:** Run your chosen solver for tolerances 10^{-2}, 10^{-4}, and 10^{-6}. Report the number of iterations required to reach each tolerance.
- **Preconditioning:** Suggest and apply a suitable preconditioner. Repeat the same experiments with the chosen preconditioner and report the number of iterations (and optionally, the CPU time) required for each tolerance.
- **Discussion:** Compare the solver performance **with and without preconditioning**. Comment on whether the preconditioner improves **accuracy** or **convergence speed**. Compute the preconditioner P using different **drop tolerances**, and fill out the following table to aid your comparison.

Table 1: Solver performance comparison with different preconditioner drop tolerances.

Drop Tolerance	$\|A - P\|_F / \|A\|_F$	**Iterations to Reach Tolerance**	$\text{cond}(P^{-1}A)$
$1e^{-1}$			
$1e^{-4}$			
$1e^{-8}$			
$1e^{-12}$			
0 (exact)			

Exercise 3. Investigate how the **Arnoldi iteration** approximates its eigenvalues (Ritz values) as the dimension of the Krylov subspace increases.

1. **Matrix Construction.** Choose the dimension of the problem k to construct the desired matrix A of dimension $2k \times 2k$ with the provided function `Matrix_ex_3(k)`.

2. **Numerical Experiment.** For the values

$$m \in \{10, 20, \ldots, k\},$$

run the Arnoldi iteration and visualize the computed Ritz values. Use the provided function `eigplot(lamt,lam,res,title)` to visualize the comparison between the true eigenvalues of A (`lamt`) and the computed Ritz values (`lam`) by the residual magnitudes (`res`).

3. **Discussion.** Comment on how the Ritz values evolve as m increases. Do they tend to approximate the eigenvalues of A more accurately? How do the residual norms behave?